

Chapter 16: Subdifferentiability of Convex Functions

Convex Analysis and Monotone Operator Theory in Hilbert Spaces

Based on Bauschke & Combettes (2017)

Chapter Overview

- **16.1:** Basic properties of subdifferentials
- **16.2:** Convex functions and subdifferentials
- **16.3:** Lower semicontinuous convex functions
- **16.4:** Subdifferential calculus (chain rule, sum rule, composition)
- **16.5:** Subdifferential of composition and coderivatives

Key Concepts:

- Subgradients and supporting hyperplanes
- Fermat's rule and optimality
- Proximal operators and monotonicity
- Calculus rules for subdifferentials

Subgradients: Definition and Geometry

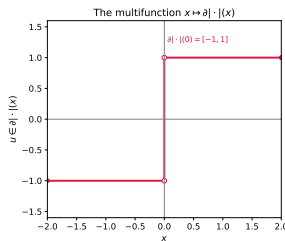
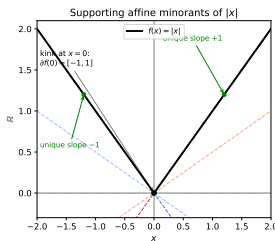
Definition 16.1: Let $f : H \rightarrow \overline{\mathbb{R}}$ be proper. A vector $u \in H$ is a **subgradient** of f at $x \in \text{dom } f$ if:

$$(\forall y \in H) \quad \langle y - x \mid u \rangle + f(x) \leq f(y)$$

The **subdifferential** $\partial f(x)$ is the set of all subgradients:

$$\partial f(x) = \{u \in H \mid (\forall y \in H) \langle y - x \mid u \rangle + f(x) \leq f(y)\}$$

Geometric Interpretation: u is the “slope” of a supporting hyperplane to the epigraph of f that touches the graph at $(x, f(x))$.



Fermat's Rule: First-Order Optimality

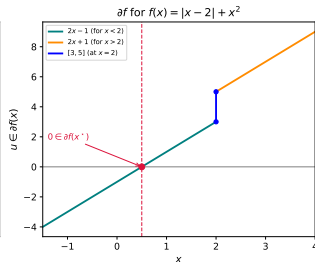
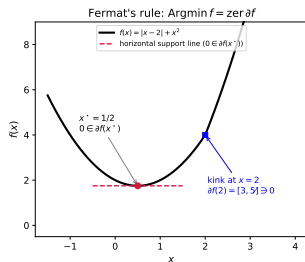
Theorem 16.3 (Fermat's Rule): Let $f : H \rightarrow \overline{\mathbb{R}}$ be proper. Then:

$$\operatorname{Argmin} f = \operatorname{zer} \partial f = \{x \in H \mid 0 \in \partial f(x)\}$$

Proof Sketch:

- $x \in \operatorname{Argmin} f \Leftrightarrow (\forall y \in H) \langle y - x \mid 0 \rangle + f(x) \leq f(y)$
- This is equivalent to $0 \in \partial f(x)$

Key Consequence: Finding a minimizer is equivalent to finding a point where 0 is a subgradient!



Example: Computing Subdifferentials

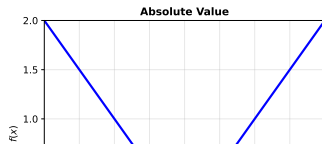
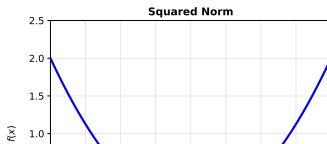
Example 16.12: For $f(x) = \frac{1}{2}\|x\|^2$, we have $\partial f = \text{Id}$ (the identity operator).

Example 16.15: For $f(\xi) = |\xi|$ in $\mathbb{R}$:

$$\partial|\xi| = \begin{cases} \{-1\}, & \text{if } \xi < 0 \\ [-1, 1], & \text{if } \xi = 0 \\ \{1\}, & \text{if } \xi > 0 \end{cases}$$

Example 16.32: For the norm $f(x) = \|x\|$:

$$\partial\|x\| = \begin{cases} \left\{ \frac{x}{\|x\|} \right\}, & \text{if } x \neq 0 \\ B(0; 1), & \text{if } x = 0 \end{cases}$$



Basic Properties of Subdifferentials I

Proposition 16.4: Let $f : H \rightarrow \overline{\mathbb{R}}$ be proper and $x \in \text{dom} f$. Then:

- (i) $\text{dom} \partial f \subseteq \text{dom} f$
- (ii) $\partial f(x) = \bigcap_{y \in \text{dom} f} \{u \in H \mid \langle y - x \mid u \rangle \leq f(y) - f(x)\}$
- (iii) $\partial f(x)$ is closed and convex
- (iv) If $x \in \text{dom} \partial f$, then f is lower semicontinuous and weakly lower semicontinuous at x

Proposition 16.5 (Connection to Conjugate): Let $f : H \rightarrow \overline{\mathbb{R}}$ be proper. Then:

$$f^{**}(x) = f(x) \quad \text{and} \quad \partial f^{**}(x) = \partial f(x)$$

Basic Properties of Subdifferentials II

Proposition 16.6 (Scaling and Composition): Let $\lambda \in \mathbb{R}_{++}$, $L \in \mathcal{B}(H, K)$, and $g : K \rightarrow \overline{\mathbb{R}}$ proper. Then:

- (i) $\partial(\lambda f) = \lambda \partial f$
- (ii) If $\text{dom}(g \circ L) \cap \text{dom} f \neq \emptyset$, then $\partial(f + g \circ L) = \partial f + L^* \circ (\partial g) \circ L$

Basic Properties of Subdifferentials III

Proposition 16.7 (Subdifferential on Product Spaces):

Let $H = \bigoplus_{i \in I} H_i$ and $f : H \rightarrow \overline{\mathbb{R}}$ proper. For $\mathbf{x} = (x_i)_{i \in I}$:

$$\partial f(\mathbf{x}) \subseteq \times_{i \in I} \partial(f \circ R_i)(x_i)$$

where $R_i : H_i \rightarrow H$ is the natural embedding.

Subdifferentials of Convex Functions

Remark 16.2: For $f \in \Gamma_0(H)$, we have $\text{dom} \partial f \neq \emptyset$ (Theorem 9.20 + Corollary 16.39).

Proposition 16.16: Let $f : H \rightarrow \overline{\mathbb{R}}$ be proper and convex. Then:

$$u \in \partial f(x) \Leftrightarrow (u, -1) \in N_{\text{epi } f}(x, f(x)) \Leftrightarrow f(x) + f^*(u) = \langle x \mid u \rangle$$

This last equality is the **Fenchel-Young equality**.

Proposition 16.17: Let $f : H \rightarrow \overline{\mathbb{R}}$ be proper and convex, $x \in \text{dom } f$. Then:

- (i) If $\text{int}(\text{dom } f) \neq \emptyset$ and $x \in \text{bdry}(\text{dom } f)$, then $\partial f(x)$ is empty or unbounded
- (ii) If $x \in \text{cont } f$, then $\partial f(x)$ is nonempty and weakly compact
- (iii) If $x \in \text{cont } f$, then $\exists \rho \in \mathbb{R}_{++}$ such that $\partial f(B(x; \rho))$ is bounded
- (iv) If $\text{int}(\text{dom } f) \neq \emptyset$, then $\text{int}(\text{dom } f) \subseteq \text{dom } \partial f$

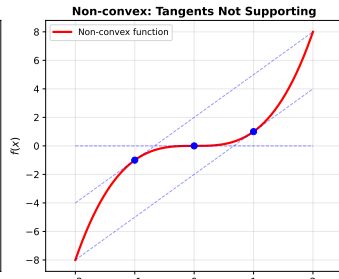
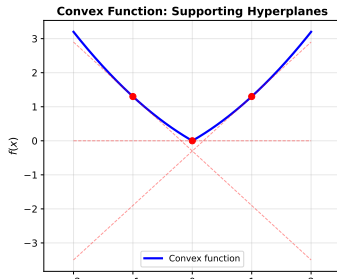
Finite-Dimensional Results

Corollary 16.18: When H is finite-dimensional and $f : H \rightarrow \overline{\mathbb{R}}$ is proper and convex:

- (i) $\emptyset \neq \text{ri}(\text{dom} f) \subseteq \text{dom} \partial f$
- (ii) f possesses a continuous affine minorant

Corollary 16.21: In finite dimensions, for $f \in \Gamma_0(H)$, the following are equivalent:

- f is supercoercive
- $(\forall u \in H) f(\langle \cdot | u \rangle)$ is coercive
- $\text{dom} f^* = H$



Lower Semicontinuous Convex Functions I

Definition: A function $f : H \rightarrow \overline{\mathbb{R}}$ is in $\Gamma_0(H)$ if it is proper, convex, and lower semicontinuous.

Proposition 16.24: If $f \in \Gamma_0(H)$ is positively homogeneous, then $f = \sigma_C$ where $C = \partial f(0)$.

Proposition 16.25: If $f \in \Gamma_0(H)$ with $\text{int}(\text{dom} f) \neq \emptyset$, then:

$$f = (f + \iota_{\text{int}(\text{dom} f)})^{**}$$

Lower Semicontinuous Convex Functions II

Proposition 16.27: For $f \in \Gamma_0(H)$:

$$\text{int}(\text{dom} f) = \text{cont} f \subseteq \text{dom} \partial f \subseteq \text{dom} f$$

Remark 16.28: The set $\text{dom} \partial f$ may not be convex! Example: $f(\xi_1, \xi_2) = \max\{g(\xi_1), |\xi_2|\}$ in $\mathbb{R}^2$.

Lower Semicontinuous Convex Functions III

Theorem 16.29 (Characterizations): For $f \in \Gamma_0(H)$, $x \in H$, $u \in H$, the following are equivalent:

- (i) $(x, u) \in \text{gra} \partial f$
- (ii) $(u, -1) \in N_{\text{epi } f}(x, f(x))$
- (iii) $f(x) + f^*(u) = \langle x \mid u \rangle$
- (iv) $(u, x) \in \text{gra} \partial f^*$

Corollary 16.30: For $f \in \Gamma_0(H)$:

$$(\partial f)^{-1} = \partial f^*$$

Chain Rule for Subdifferentials I

Proposition 16.42 (General Chain Rule): Let $L \in \mathcal{B}(H, K)$, $f \in \Gamma_0(H)$, $g \in \Gamma_0(K)$, with $L(\operatorname{dom} f) \cap \operatorname{dom} g \neq \emptyset$. If $(f + g \circ L)^* = f^* \boxplus (L^* \circ g^*)$, then:

$$\partial(f + g \circ L) = \partial f + L^* \circ (\partial g) \circ L$$

Chain Rule for Subdifferentials II

Example 16.43: Let $f \in \Gamma_0(H)$ and $\gamma \in \mathbb{R}_{++}$. Then:

$$\partial(f + (\gamma/2)\|\cdot\|^2) = \partial f + \gamma \cdot \text{Id}$$

Proposition 16.44 (Proximal Operator): For $f \in \Gamma_0(H)$ and $x, p \in H$:

$$p = \text{prox}_f x \Leftrightarrow x - p \in \partial f(p)$$

Equivalently:

$$\text{prox}_f = (\text{Id} + \partial f)^{-1}$$

Proposition 16.45: For $f \in \Gamma_0(H)$:

$$\text{ran}(\text{Id} + \partial f) = H$$

This means the proximal operator is everywhere defined and single-valued.

Chain Rule for Subdifferentials III

Remark 16.46 (Sum Rule Exactness): Let $f, g \in \Gamma_0(H)$ with $\operatorname{dom} f \cap \operatorname{dom} g \neq \emptyset$. Then:

$$f^* \boxplus g^* \text{ is exact on } \operatorname{dom}(f + g)^* \Rightarrow \partial(f + g) = \partial f + \partial g$$

Conversely:

$$\partial(f + g) = \partial f + \partial g \Rightarrow f^* \boxplus g^* \text{ is exact on } \operatorname{dom}(f + g)^*$$

But the implication $\partial(f + g) = \partial f + \partial g \Rightarrow (f + g)^* = f^* \boxplus g^*$ can fail!

Subdifferential of Sum I

Theorem 16.47 (Sum Rule): Let K be a Hilbert space, $f \in \Gamma_0(H)$, $g \in \Gamma_0(K)$, $L \in \mathcal{B}(H, K)$.

If one of these holds:

- (i) $0 \in \text{sri}(\text{dom}g - L(\text{dom}f))$
- (ii) K finite-dimensional, g polyhedral, $\text{dom}g \cap L(\text{dom}f) \neq \emptyset$
- (iii) H, K finite-dimensional, f, g polyhedral, $\text{dom}g \cap L(\text{dom}f) \neq \emptyset$

Then: $\partial(f + g \circ L) = \partial f + L^* \circ (\partial g) \circ L$

Corollary 16.48 (Simple Cases): For $f, g \in \Gamma_0(H)$, if one holds:

- (i) $0 \in \text{sri}(\text{dom}f - \text{dom}g)$
- (ii) $\text{dom}f \cap \text{int}(\text{dom}g) \neq \emptyset$
- (iii) $\text{dom}g = H$
- (iv) H finite-dimensional and $\text{ri}(\text{dom}f) \cap \text{ri}(\text{dom}g) \neq \emptyset$

Then: $\partial(f + g) = \partial f + \partial g$

Subdifferential of Sum III

Corollary 16.50 (Multiple Functions): For $m \geq 2$ functions $f_i \in \Gamma_0(H)$, if one condition holds, then:

$$\partial \left(\sum_{i \in I} f_i \right) = \sum_{i \in I} \partial f_i$$

Example 16.51: The conditions are necessary! Example in $\mathbb{R}^2$ where $\partial(\iota_C + \iota_D) \neq \partial\iota_C + \partial\iota_D$ when $C \cap D = \{(0, 0)\}$ only.

Mean Value Theorem I

Theorem 16.56 (Mean Value): Let $f \in \Gamma_0(H)$, $x_0, x_1 \in \operatorname{dom} f$. Suppose one holds:

- (i) $0 \in \operatorname{sri}(\operatorname{dom} f - \operatorname{aff}\{x_0, x_1\})$
- (ii) $\operatorname{aff}\{x_0, x_1\} \cap \operatorname{int}(\operatorname{dom} f) \neq \emptyset$
- (iii) H finite-dimensional, $\operatorname{aff}\{x_0, x_1\} \cap \operatorname{ri}(\operatorname{dom} f) \neq \emptyset$

Then: $f(x_1) - f(x_0) \in \langle x_1 - x_0 \mid \partial f(\cdot)_{x_0, x_1}(\cdot) \rangle$

Mean Value Theorem II

Corollary 16.57 (Lipschitz Continuity): If $f : H \rightarrow \mathbb{R}$ is convex and continuous, with $\beta = \sup \|\text{ran} \partial f\| < \infty$, then f is Lipschitz continuous with constant β .

Theorem 16.58 (Brøndsted-Rockafellar): For $f \in \Gamma_0(H)$, $(y, v) \in \text{dom}(f \oplus f^*)$, $\lambda, \mu \in \mathbb{R}_+$ with $f(y) + f^*(v) \leq \langle y \mid v \rangle + \lambda\mu$, there exist $(z, w) \in \text{gra} \partial f$ with $\|z - y\| \leq \lambda$ and $\|w - v\| \leq \mu$.

This is a fundamental result in monotone operator theory!

Subdifferential of Composition I

Definition 16.67 (Coderivative): Let K be Hilbert, $F : H \rightarrow 2^K$ with $\text{gra}F$ nonempty and convex. The **coderivative** of F at $(x, y) \in \text{gra}F$ is:

$$D^*F(x, y) : K \rightarrow 2^H : v \mapsto \{u \in H \mid (u, -v) \in N_{\text{gra}F}(x, y)\}$$

Example 16.68: If $F = \partial f$ for $f : H \rightarrow \overline{\mathbb{R}}$ proper and convex, then for $(x, u) \in \text{gra}\partial f$:

$$D^*(\partial f)(x, u)(v) = \partial f^*(u, -v)$$

Subdifferential of Composition II

Proposition 16.69: Let $f : H \rightarrow \overline{\mathbb{R}}$ be proper and convex, and $F(x) = \{y \in K \mid f(x) \leq y\}$. For $x \in \text{dom} f$:

$$D^*F(x, f(x))(v) = \begin{cases} v\partial f(x), & \text{if } v > 0 \\ N_{\text{dom} f}x, & \text{if } v = 0 \\ \emptyset, & \text{if } v < 0 \end{cases}$$

Subdifferential of Composition III

Theorem 16.71 (Composition Rule): Let K Hilbert, $g \in \Gamma_0(K)$, $F : H \rightarrow 2^K$ with $\text{gra}F$ closed and convex. For proper $f(x) = \inf\{g(y) \mid y \in F(x)\}$, if f proper, $\bar{x} \in \text{dom}f$, $\bar{y} \in F(\bar{x})$ with $f(\bar{x}) = g(\bar{y})$, and $(0, 0) \in \text{sri}(\text{gra}F - \text{dom}g)$, then:

$$\partial f(\bar{x}) = \bigcup_{\substack{(u,v) \in \partial g(\bar{y}) \\ g(\bar{y}) = f(\bar{x})}} u + D^*F(\bar{x}, \bar{y})(v)$$

Subdifferential of Composition IV

Corollary 16.72 (Composition with Increasing Function): For $f : H \rightarrow \mathbb{R}$ continuous convex, $\phi \in \Gamma_0(\mathbb{R})$ increasing on $\text{ran} f$, if $(\text{ri}(\text{ran} f) + \mathbb{R}_{++}) \cap \text{ri}(\text{dom} \phi) \neq \emptyset$, then:

$$\partial(\phi \circ f)(\bar{x}) = \{\alpha u \mid \alpha \in \partial\phi(f(\bar{x})), u \in \partial f(\bar{x})\}$$

Application: Proximal Gradient Method

Practical Importance: The chain rule is crucial for optimization algorithms.

```
import numpy as np

# Proximal gradient method for  $f(x) + g(x)$ 
# where  $f$  is smooth and  $g$  has an easy proximal operator

def proximal_gradient(x0, f_grad, prox_g, step_size, max_iters=1000, tol=1e-4):
    x = x0
    for k in range(max_iters):
        # Gradient step on  $f$ 
        grad_f = f_grad(x)
        x_grad = x - step_size * grad_f

        # Proximal step on  $g$ 
        x_new = prox_g(x_grad, step_size)

        # Check convergence
        if np.linalg.norm(x_new - x) < tol:
            break
        x = x_new
    return x

# Example: Lasso problem: minimize  $0.5||Ax - b||^2 + \lambda||x||_1$ 
#  $f(x) = 0.5||Ax - b||^2$ ,  $g(x) = \lambda||x||_1$ 
```

Key Applications and Consequences

1. Optimization:

- Fermat's rule: x^* minimizes $f \Leftrightarrow 0 \in \partial f(x^*)$
- First-order optimality condition for non-smooth functions
- Proximal operator is a building block for many algorithms

2. Monotone Operators:

- ∂f is a monotone operator: $\langle u_1 - u_2, x_1 - x_2 \rangle \geq 0$ for $(x_i, u_i) \in \text{gra} \partial f$
- Subdifferential is maximal monotone (important for existence results)
- Brøndsted-Rockafellar theorem relates approximate solutions

3. Convex Analysis:

- Duality via Fenchel-Young equality: $f(x) + f^*(u) = \langle x \mid u \rangle \Leftrightarrow u \in \partial f(x)$
- Determines whether functions are lower semicontinuous
- Characterizes inf-projections and integral functions

4. Algorithms:

- Forward-backward splitting (proximal gradient method)
- Douglas-Rachford splitting
- Alternating direction method of multipliers (ADMM)

Numerical Example: Soft Thresholding

Problem: Solve $\min_x \frac{1}{2} \|Ax - b\|^2 + \lambda \|x\|_1$ (Lasso)

Proximal Operator of ℓ^1 -norm:

$$\text{prox}_{\lambda \|\cdot\|_1}(y) = \text{sign}(y) \odot \max\{|y| - \lambda, 0\}$$

(Soft thresholding: shrink coefficients toward zero)

```
def soft_threshold(y, threshold):  
    """Soft thresholding operator (proximal of L1 norm)"""  
    return np.sign(y) * np.maximum(np.abs(y) - threshold, 0)  
  
# This comes from the subdifferential:  
# u in partial(lambda*||x||_1) iff  
#   u_i = sign(x_i)*lambda if x_i != 0  
#   u_i in [-lambda, lambda] if x_i = 0
```

The subdifferential tells us exactly what the proximal operator should do!

Chapter Summary

Key Definitions:

- Subgradient: generalizes gradient to non-smooth functions
- Subdifferential: set of all subgradients at a point
- Proper, convex, lower semicontinuous functions: $\Gamma_0(H)$

Fundamental Theorems:

- Fermat's Rule: optimality via $0 \in \partial f(x^*)$
- Fenchel-Young: $f(x) + f^*(u) = \langle x | u \rangle \Leftrightarrow u \in \partial f(x)$
- Brøndsted-Rockafellar: approximate solutions exist nearby
- Composition Rule: calculus for subdifferentials of composed functions

Calculus Rules:

- Scaling: $\partial(\lambda f) = \lambda \partial f$
- Sum rule: conditions for $\partial(f + g) = \partial f + \partial g$
- Chain rule: $\partial(f + g \circ L) = \partial f + L^* \circ (\partial g) \circ L$
- Proximal operator: $\text{prox}_f = (\text{Id} + \partial f)^{-1}$

Next Steps: These tools are essential for monotone operator theory and splitting methods!

Connections to Other Topics

Chapter 13: Conjugate functions

- Fenchel-Young equality relates conjugate to subdifferential
- ∂f and ∂f^* are inverses of each other

Chapters 6-9: Convexity and closedness

- Lower semicontinuity ensures non-empty subdifferential (Chapter 9)
- Subdifferential characterizes closed convex sets (indicator functions)

Chapter 17: Monotone operators

- ∂f is the canonical example of a monotone operator
- Maximality of monotone operators
- Resolvent and inverse operators

Chapters 23-26: Splitting methods

- Forward-backward splitting uses prox_f
- Douglas-Rachford uses monotone operators
- Three-operator splitting combines multiple subdifferentials